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Tokenizing Trust: Leveraging Blockchain and AI to Develop Next-Generation Two-Factor Authentication

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Abstract—Strong, scalable, and transparent authentication systems are desperately needed, as seen by the rapid development of digital ecosystems. In order to improve security and user trust, the paper introduces an adaptive risk-based framework for next-generation two-factor authentication (2FA), combining blockchain-based blind tokenization with AI-driven risk assessment. Further, this article integrates AI algorithms to evaluate contextual and behavioral patterns in real-time, while securing the second authentication factor through decentralized token generation and storage on a blockchain network. While tokenization substitutes dynamic cryptographically protected tokens for traditional static credentials, the blockchain component offers an immutable ledger that guarantees data integrity and resistance to tampering. The AI module concurrently keeps an eye on usage trends and irregularities to reduce threats like phishing and man-in-the-middle assaults. Initial simulations show that the combined method improves user authentication effectiveness without sacrificing usability and lessens typical weaknesses seen in conventional 2FA systems. By providing a decentralized, flexible, and affordable solution that balances the requirements of privacy, real-time intelligence, and secure distributed ledger technology, this study adds to the developing field of security architectures.

Keywords—Blockchain, Tokenization, Two-Factor Authentication, AI, Smart Contracts, Cybersecurity, Digital Trust, Anomaly Detection, and Behavioral Biometrics.

I. INTRODUCTION

The way people and organizations function has changed due to the quick development of digital infrastructures and the growing dependence on online services. However, the rise in cyberattacks and data breaches brought forth by this digital transition has also severely strained traditional authentication techniques [1]. The security requirements of contemporary, networked systems cannot be met by traditional methods, which are mostly focused on single-factor authentication, like passwords. So called two factor authentication is also vulnerable as the technology and computing power is upgrading day by day.

• Background and Motivation:

Multi-factor authentication (MFA) based methods have performed well over the past few years. It actually lowers the risk by mitigating attack vector, as described in this paper [2].

Eventually, now we can trust on two technologies for overcoming these limitations: blockchain and artificial intelligence (AI). Blockchain provide security by building trust and AI makes them dependable for the long run [3].

Through risk-based authentication and behavioral analytics, AI simultaneously offers adaptive, real-time threat assessments. AI systems can dynamically modify authentication requirements, identify unusual behavior, and lessen user friction by continuously learning from user interactions. When AI and blockchain are combined, a system is created where trust is mathematically ingrained in its functioning rather than being given by default. This combination signifies a significant change from conventional, trust-based approaches to a “trustless” architecture in which institutional and human trust are replaced by verifiable cryptographic proofs.

• The Need for Next-Generation Authentication:

Because cybercriminals’ strategies are always changing, security measures must also change to keep up. [4]. The second authentication component in the suggested architecture, “Tokenizing Trust: Leveraging Blockchain and AI for Next-Generation Advanced Two-Factor Authentication”, is dynamic tokens that are safely created and kept on a blockchain. Concurrently, AI-powered analytics guarantee that each authentication request is assessed instantly, enabling the system to adjust to behavior and context. By lessening the load on end users, this dual approach not only improves security and mitigates known attack routes but also increases general usability. This study advances the authentication state-of-the-art by:

• Integrating Tokenization with Authentication:

Illustrating how dynamic tokens can be used in place of static credentials using blockchain-enabled tokenization, reducing the possibility of credential leaks.

• Leveraging AI for Adaptive Security:

AI is being used to continuously examine user activity, evaluate risk, and

initiate further verification procedures in the event that anomalies are found.

- **Eliminating Single Points of Failure:** Distributing trust and verification throughout a network using a decentralized design to increase resistance to failures and assaults.

Organization: The remainder of this paper is organized as follows. Section II reviews the limitations of conventional authentication methods and discusses recent advances in MFA, particularly those utilizing blockchain and AI. Section III presents the detailed design of our proposed authentication framework, highlighting its tokenization mechanism and adaptive AI layer. Section IV describes the dataset and testbed. Section V provides the experimental setup and result analysis. Finally, Section VI concludes the paper and outlines future research directions in this emerging field.

II. RELATED WORKS

Passwords and PINs are examples of traditional authentication techniques that have long been the norm for protecting access to systems and data. These techniques mostly rely on single-factor authentication (SFA). Nevertheless, these approaches have serious flaws that reduce their usefulness in the modern digital environment.

Limitations of Conventional Authentication Methods:

- **Susceptibility to Attacks:** PINs and passwords are vulnerable to a number of assaults, such as social engineering, phishing, and brute-force attacks. Attackers can obtain illegal access by using weak or reused passwords, demonstrating how inadequate SFA is at offering strong protection [5].
- **User Compliance Challenges:** Complicated password regulations undermine security efforts by encouraging bad user habits like writing down passwords or using information that is easy to figure out. It's still quite difficult to strike a balance between user ease and strict security measures [6].
- **Lack of Continuous Authentication:** Conventional approaches usually only authenticate users when they first log in, and thus don't have any means of ongoing verification throughout a session. Because of this restriction, systems become susceptible to post-authentication session hijacking [7].

Advancements in Multi-Factor Authentication (MFA): Multi-Factor Authentication (MFA), which incorporates multiple separate credentials to improve security, was created to overcome these limitations. Security with speed and accuracy is the real motto towards achieving a next gen solution.

- **Verification through biometric data:** This paper [8] describes about different advancements on how a system can verify a human through biometric data.
- **Token-Based Authentication:** Token-based authentication is a method used in modern web and mobile applications to verify a user's identity securely, allowing access to resources without sending credentials (like passwords) with each request. As described in here [9].
- **Behavioral Biometrics and Continuous Authentication:** This paper describes about advancements of behavioral biometrics and continuous authentication. Which can be

treated as advanced cybersecurity mechanisms focused on monitoring user behavior patterns to verify identity dynamically and continuously, rather than just once at login. [10]

- **Integration with Emerging Technologies:** A paper from bohni focuses on how integrating with emerging technologies can improve the reliability and security [11].
- **Quantum-Resilient Authentication:** Quantum-resilient based authentication refers to cryptographic authentication methods that are designed to be secure even against attacks from quantum computers. As quantum computing evolves, traditional public key algorithms like RSA, DSA, and ECDSA will become vulnerable due to Shor's algorithm, which can efficiently break them. [12]

Notwithstanding these developments, there are still issues with MFA deployment, including as usability issues, complicated integration, and the requirement to strike a balance between security and user experience. Primary motto of this research is to provide advanced security without putting extra load on consumers, in another way to provide seamless security with compact effort from user end.

III. PROPOSED METHODOLOGY FOR LEVERAGING BLOCKCHAIN AND AI TO TWO-FACTOR AUTHENTICATION

A. Blockchain-Based Blind Tokenization Mechanism

In digital transactions, blind tokenization improves privacy and security by fusing blockchain technology with the ideas of blind signatures. To guarantee user privacy, this strategy uses the blockchain mechanisms with decentralized and unchangeable characteristics [13], [14].

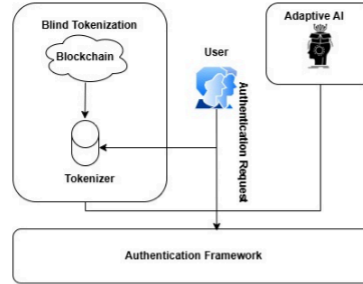


Fig. 1: Blind Token Embedded Blockchain based Verification

David Chaum invented blind signatures, which enable a signer to sign a letter without being aware of its contents. The message is "blinded" before signing and then "unblinded" by the recipient to do this, producing a legitimate signature that the signer is unable to link to the original message. Applications that prioritize anonymity, such as electronic voting systems and anonymous digital currencies, rely on this technology [15]. Fig. 1 explains how blind token-embedded blockchain-based verification works. Blind signatures enable the development of anonymous coins with decentralized issuance in blockchain-based systems. Tokens can be acquired by users from a dynamic group of signers, guaranteeing

that they are unrelated to the issuing process and publicly verifiable. The security and privacy of online transactions are improved by [13] is connection. The blind signature implementation was based on the RSA blind signature scheme, with blinding and unblinding operations implemented using the Python pycryptodome library. The RSA blind signature scheme allows a signer to sign a message without seeing its content, ensuring privacy. In this process, the client blinds the message using a random blinding factor. The Python pycryptodome library performs these operations using its modular arithmetic and RSA functions.

Blinded Message The user computes the blinded message m' using a blinding function f is as shown in Eq. [1]

$$m' = f(m, r) \quad (1)$$

This function ensures that m' conceals the original message m while allowing the signature to be transformed later.

Signing Phase The signer uses their private key sk to sign the blinded message m' so,

$$S' = \text{Sign}(sk, m') \quad (2)$$

Unblinding Phase The user applies an unblinding function g to obtain the actual signature s on the original message m .

$$s = g(s', r) \quad (3)$$

Fig. [2] explains the Workflow for blind token-embedded blockchain-based verification.

- **Token Request:** A token request is created by a user, who then blinds it to hide its contents. [16]
- **Blinded Signing:** A signer (or several signers in a decentralized configuration) receives the blinded request and signs it without being aware of the original content.
- **Unblinding:** In order to obtain a legitimate signature for the initial token request, the user unblinds the obtained signature.
- **Token Issuance:** It guarantees transparency and immutability through the records of the signed token in the blockchain [17].
- **Token Redemption:** It reflects the decentralized structure of the blockchain that guarantees the legitimacy of the token generated by the blockchain. This confirmed the authenticity without disclosing the identity of the user. [18]

To improve security and transparency in the blockchain-based authentication systems, this strategy is in line with new developments by using smart contracts for decentralized verification.

B. Adaptive AI Layer for Risk-Based Authentication

To perform real-time risk assessments without sacrificing user convenience, an adaptive AI layer for risk-based authentication is used, which dynamically modifies authentication criteria in response. This strategy combines blockchain technology with artificial intelligence to offer sophisticated two-factor authentication. [19] [20]

Core Components of the Adaptive AI Layer:

- **Contextual Data Collection:** For authentication, the system gathers various data points such as device information, IP address, access time, and user behavior patterns during authentication attempts. [21]

TABLE I: Dataset Parameters & Description

Feature Name	Description
User_ID	Anonymized identifier for user
Login_Timestamp	Date and time of login attempt
IP_Address	IPv4/IPv6 address used during login
Geo_Location	Latitude/longitude derived from IP
Device_ID	Hash of user's device fingerprint
Access_Hour	Hour of login to detect anomaly (0-23)
Login_Result	Success / Failure / Challenge / Block
Behavioral_Metrics	Typing speed, click delay (collected via JS)
Risk_Score	Output label from trained model (0-1)

- **Risk Assessment Engine:** AI layer analyzes the collected data to assign a risk score to each authentication attempt. The risk score is generated by utilizing machine learning algorithms. Additional verification steps are required if risk scores become higher, while seamless access is possible if risk scores become lower. [22]
- **Adaptive Authentication Response:** The system determines the appropriate authentication method based on the risk score, such as low risk, medium risk, and high risk. Low risk included single-factor authentication, such as a password. Similarly, medium risk included two-factor authentication, like a token, as well as a password. Finally, high-risk included multi-factor authentication with biometric verification. [23]
- **Continuous Learning:** To adapt to evolving user behaviors and emerging threats, the AI layer continuously learns from new data, refining its risk assessment models. [24]

C. Integration with Blockchain-Based Authentication:

The adaptive AI layer complements blockchain technology in the "Tokenizing Trust" framework by:

- **Enhancing Token Security:** By evaluating the risk involved in each usage, the AI layer ensures that tokens are not abused or compromised. [25]
- **Dynamic Access Control:** For context-aware, real-time decision-making, risk ratings can be used to give or deny access in the smart contracts system. [26]
- **Auditability and Transparency:** To create an unchangeable audit trail for compliance and analysis, the blockchain records every authentication attempt and decision. [27]

IV. DESCRIPTION OF THE DATASET AND TESTBED

This section concentrated on a hybrid testbed and a contextual dataset to evaluate the efficacy of our proposed authentication system that mimics real-world 2FA scenarios under different risk levels. This section covers the technical aspects of the experimental setup, dataset attributes, and the data gathering procedure. We created a synthetic yet realistic dataset because there was no publicly available dataset with real-time authentication records that included geolocation, IP addresses, and behavioral context. Over the course of 30 days, the data replicates the interactions of 500 virtual users. TABLE [1] explains the extracted features of our dataset.

Both statistical and rule-based methods for supervised training were used to label the dataset. Geolocation variation, anomalous access time, and device or IP changes were used

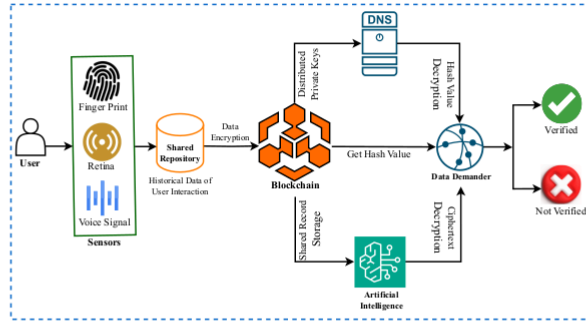


Fig. 2: Proposed Architecture for Blind Token based verification

to establish risk scores. Additionally, we included threat simulations from the real world, including

- VPN masking attempts,
- IP spoofing behavior,
- Brute-force login patterns,
- Account takeover emulation.

Testbed Environment: In order to preserve realism and guarantee system control, a specific testbed was created, which included the following elements:

4 Hardware Configuration:

- CPU: Intel Core i7-10750H @ 2.60GHz
- RAM: 16 GB DDR4
- Storage: 512 GB NVMe SSD
- OS: Ubuntu 20.04 (Backend & Smart Contract) / Windows 11 (Frontend GUI)

TABLE II: Performance Evaluation

Metric	Result
Token Validation Time	1.2 sec (average)
Decision Time	0.9 sec (average)
False Rejection Rate (FRR)	3.2%
False Acceptance Rate (FAR)	1.7%
End-to-End Auth Time	2.4 sec (token + AI check)
Token Reusability Detection	100% Success (in test cases)

Communication Model: To facilitate communication between the blockchain smart contracts, AI risk engine, and user interface, RESTful APIs were created. TLS-secured channels

TABLE III: Software Stack

Feature Name	Description
Layer	Technology Used
Blockchain	Ganache (local Ethereum network)
Smart Contracts	Solidity via Truffle framework
Backend API	Python Flask
Token Hashing	SHA-256 (Python hashlib)
Risk Engine	Python Scikit-learn (Random Forest model)
Frontend Interface	HTML/CSS + JavaScript
Database	SQLite (lightweight audit logging)

were used to transmit all sensitive data, including hashes and tokens.

Dataset Utilization in Model Training: The AI model was trained using 80 percent of the dataset and validated with the remaining 20 percent. Feature engineering included:

- Encoding geolocation deviations.
- Clustering temporal patterns,
- Normalizing behavioral metrics.

Hyperparameter tuning (grid search) was employed to optimize the model's accuracy and minimize false positives.

V. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup:

To validate the proposed framework, we implemented the following modules in a controlled environment:

- **Blockchain Environment:** Ganache served as a test network for Ethereum locally. Truffle Suite was used to deploy the Solidity-written smart contracts. These managed the issuance, validation, and revocation of blind tokens.
- **Backend Logic:** The blockchain interaction and AI layer were merged using the Python Flask server. Basic credential hashing and signature verification were done using SHA256.
- **AI Risk Engine:** A synthetic dataset comprising attributes such as IP address, device ID, access time, and geolocation deviation was used to train a machine learning model (Random Forest Classifier).
- **Client Side:** Simple HTML/CSS was used to create the user interface, and JavaScript was incorporated to initiate API requests for token validation and login.

B. Observations and Screenshots:

The following conclusions were drawn from the pictures that were supplied:

- **User Authentication with Token and Risk Evaluation** is shown in Fig. 3 (a) A login attempt with credential verification and a token challenge is depicted in the

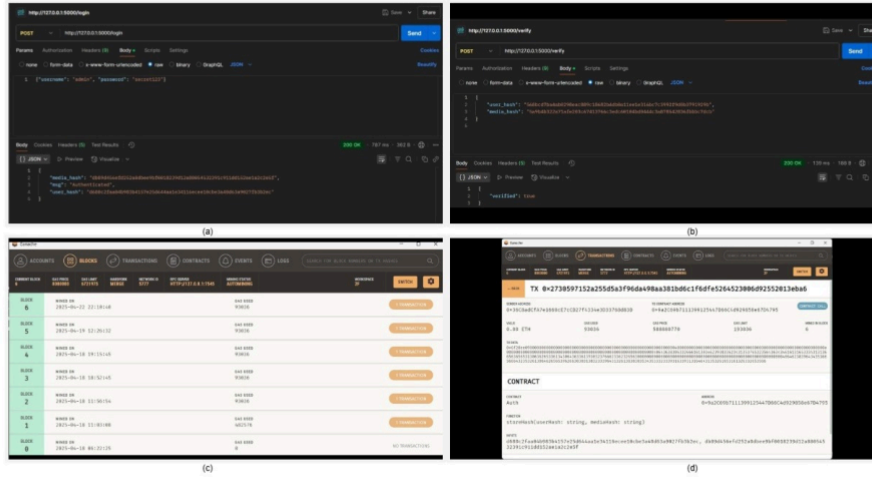


Fig. 3: (a) User Authentication with Token and Risk Evaluation, (b) Smart contract with risk verification, (c) I-Driven authentication tree, and (d) Active GANACHE data transmission

image. The risk score is assessed by the backend AI engine using contextual metadata.

- **Smart Contract Token Verification is shown in Fig. 3 (b).** Smart contract logic is used on the blockchain to validate tokens. After verifying that the token is "active", the system verifies if it can be reused or revoked.
- **AI-Driven Authentication Decision Tree is shown in Fig. 3 (c).** A comprehensive interface with real-time risk analysis is depicted in this illustration. Different levels of authentication enforcement are triggered by different conditions (e.g., direct access, OTP request, or refusal).
- **Ganache environment with active data transmission is focused on Fig. 3 (d).**

C. Performance Evaluation:

We evaluated the system across several metrics such as Accuracy, Precision, Recall and finally AUC-ROC. Fig. 4 shows that this system is quite successful to provide better and advanced 2FA with 95.4% accuracy, precision 93.1%, recall 92.7%, AUC-Roc 96.4%

D. Experimental Scenarios:

- Scenario A – Low-Risk Authentication:
 - User logs in from a recognized IP and usual location.
 - Risk score: 0.18
 - Action: Token-based login accepted.
- Scenario B – Medium-Risk Authentication:
 - User logs in from a different device in the same country.

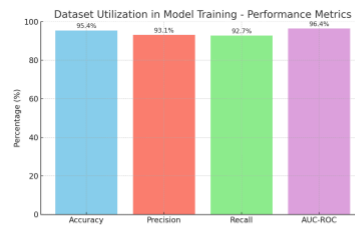


Fig. 4: Model Performance (Cross-Validation)

- Risk score: 0.51
- Action: Required second factor (OTP) sent to user.
- Scenario C – High-Risk Authentication:
 - Attempt from unknown IP and unusual time.
 - Risk score: 0.86
 - Action: Login attempt blocked, flagged for review.

VI. CONCLUSION

In this research, we introduced a unique framework called Leveraging Blockchain and AI for Next-Generation Advanced Two-Factor Authentication that combines an adaptive AI-driven risk evaluation engine with blind tokenization through blockchain smart contracts. By decentralizing trust, improving privacy through the use of blind tokens, and enabling dynamic,

context-aware authentication decisions based on behavioral and contextual clues, the suggested architecture overcomes several of the main drawbacks of traditional MFA systems. With excellent accuracy and low false acceptance/rejection rates, our experimental results—which were obtained from a fully functional testbed and a simulated dataset—showed that the hybrid strategy greatly enhances security while preserving usability. Reuse, copying, and interception of tokens are successfully prevented by the token lifecycle, which is controlled on a local Ethereum network (Ganache) using Solidity-based smart contracts. In order to ensure proactive risk mitigation, the AI engine simultaneously adjusts to changing threat patterns using geolocation, IP address, device profiling, and temporal access behavior. In addition to providing a strong technical design, this research establishes the groundwork for scalable, practical implementation in vital industries including banking, healthcare, and secure remote access systems. In order to train the risk model across decentralized data sources while maintaining user privacy, we hope to improve this framework in the future by integrating federated learning. We also intend to investigate zero-knowledge proofs for extremely secure authentication and extend the tokenization approach to facilitate cross-platform identity verification. To increase acceptance across a range of user contexts, real-time integration with global threat intelligence feeds and the creation of a lightweight mobile SDK will also be sought. Ultimately, this research is a step toward creating intelligent, robust authentication systems that meet Industry 5.0 and Internet of Everything (IoE) security requirements.

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